



UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES

B.Sc. General Degree Second Year First Semester Course Unit Examination – Sept/Oct 2025

DEPARTMENT OF MATHEMATICS

MAT 211 1.0 Linear Algebra II

Time: One hour. No. of questions: 04 No. of pages: 01 Total marks: 80

Instructions:

- Answer all questions.
- You may bring one A4 Sheet of information for the exam.
- Basic scientific calculators are allowed.
- Usual notation applies throughout.

1. (i) Write the quadratic form $3x^2 + 2xy + 3y^2$ as a product of matrices $X^T A X$, where A is a symmetric matrix. [8]
 (ii) Explain why the matrix A can be diagonalised orthogonally. [8]
 (iii) Calculate the orthonormal matrix that diagonalises A . [8]
 (iv) Hence, find the conic section of $3x^2 + 2xy + 3y^2 = 17$. [9]

2. Prove or disprove the following statements:
 - (i) Matrices A and B are similar, then the matrices A^3 and B^3 are similar. [10]
 - (ii) Let $X = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Then $X^{2025} = X$. [10]
 - (iii) If the eigenvalues of A are 2, 3 and 4, then the eigenvalues of PAP^{-1} are $1/2, 1/3$ and $1/4$. [7]
 $A^{-1} = \lambda^{-1}$

3. Suppose the matrix A has the characteristic polynomial $p(\lambda) = (\lambda + 9)^2(\lambda - 3)^2(\lambda - 1)$
 with geometric multiplicity of $\lambda = -9$ is 2, and geometric multiplicity of $\lambda = 3$ is 1.
 Without doing any calculation, determine a possible Jordan canonical form of A . [12]

4. (i) Find a basis for the subspace $W = \{(x, y, z) | x - y = 0\}$ of the vector space $\mathbb{R}^3$. [5]
 (ii) Hence, write down the dimension of the subspace W . [3]

